

# VIKAS GURJAR

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## EDUCATION

Vellore Institute of Technology (VIT)

B.Tech in Computer Science and Engineering — CGPA: 8.09

2022 – Present

Madhya Pradesh, India

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL

**AI / Machine Learning:** PyTorch, TensorFlow, Scikit-learn, OpenCV, NLTK

**Generative AI:** RAG (Retrieval-Augmented Generation), FAISS Vector DB, Google Gemini API, LangChain, Embeddings

**Backend & APIs:** FastAPI, Flask, RESTful Services, Asynchronous Programming

**MLOps & Infrastructure:** Docker, MLflow, GitHub Actions (CI/CD), Git, AWS (EC2)

**Databases & Tools:** MySQL, Redis, SQLite, Linux (Bash Scripting), Streamlit

## PROJECTS

Enterprise AI Copilot – Private Document Intelligence Platform

[Link](#)

Python, FastAPI, FAISS, Gemini, Redis, Docker

Sep 2025 – Dec 2025

- Engineered a **production-grade RAG system** to query 1,000+ enterprise documents, reducing information retrieval time by **60%**.
- Implemented **FAISS HNSW indexing** and semantic search, achieving a sub-200ms retrieval latency and a **Recall@10 of 92%**.
- Designed a modular backend with **FastAPI**, enabling persistent conversation memory via **Redis** for context-aware responses.
- Containerized the application using **Docker** and set up health monitoring, ensuring **99.9% uptime** during load testing.

Real-Time Financial Anomaly Detection Pipeline

[Link](#)

Python, Scikit-learn, MLflow, Docker, CI/CD

March 2025 – June 2025

- Developed a high-throughput ML pipeline to detect fraudulent transactions, handling **500+ transactions per second (TPS)**.
- Solved data imbalance issues using **SMOTE**, improving the model's F1-score from 0.72 to **0.89**.
- Integrated **MLflow** for experiment tracking and model registry, reducing model versioning errors by **100%**.
- Automated testing and deployment via **GitHub Actions (CI/CD)**, reducing deployment cycle time by **40%**.

Autonomous Lane Detection System (Computer Vision)

[Link](#)

Python, OpenCV, NumPy

Oct 2024 – Jan 2025

- Built a lane detection algorithm using **Canny Edge Detection** and **Hough Transform**, achieving **95% accuracy**.
- Optimized the image processing pipeline to run at **35 FPS** on CPU hardware.
- Implemented adaptive thresholding for consistent detection across varying lighting conditions.

## EXTRACURRICULAR ACTIVITIES

- BlockDAG Hackathon 2025 (Round 2 Qualifier):** Built **GhostWallet**, a decentralized privacy-focused wallet, integrating smart contracts for secure transactions.
- Open Source Contribution:** Active contributor to developer communities; engaged in optimizing code repositories and documentation.
- Volleyball Team Captain:** Represented college in inter-university tournaments; led team strategy and training sessions.

## CERTIFICATIONS

- OCI AI Foundations — Oracle University (2025)
- Generative AI using IBM Watsonx — IBM Career Education Program (2025)
- Cyber Security Analyst — IBM Career Education Program (2025)